

2.1 Checking for understanding

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POSSIBLE POINTS: 1

Find $\lim_{x \rightarrow 36} \frac{6 - \sqrt{x}}{x - 36}$. Please give answer as a fraction(no spaces), for example: -2/17 . The answer is:

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2.1 Checking for understanding

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Let f be defined as follows,

$$f(x) = \begin{cases} x^2 - 2, & \text{for } -3 < x < 3 \\ 4, & \text{for } x = 3 \\ x + 4, & \text{for } 3 < x < \infty \end{cases}$$

What is the limit of $f(x)$ as x approaches 3?

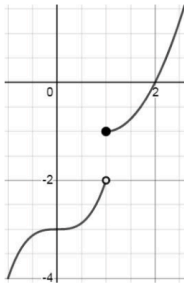
Give the answer as a number, for example: 18 or write: DNE

POSSIBLE POINTS: 1

2.1 Checking for understanding

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POSSIBLE POINTS: 3

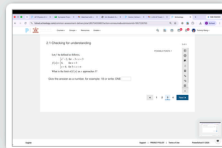


Using the graph shown, what is:

(a) $\lim_{x \rightarrow 1^-} f(x) =$

(b) $\lim_{x \rightarrow 1^+} f(x) =$

(c) $\lim_{x \rightarrow 1} f(x) =$



2.3 Checking for understanding

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What is $\lim_{x \rightarrow 2} \left(\frac{x^2}{x-2} - \frac{4}{x-2} \right)$?

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2.3 Checking for understanding

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What is $\lim_{h \rightarrow 0} \frac{5 - \sqrt{25 + h}}{h}$?

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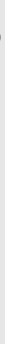
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Review

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c) $\lim_{x \rightarrow 7} f(x) =$

2.4 Checking for understanding

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POSSIBLE POINTS: 1

What is $\lim_{x \rightarrow \infty} \frac{-x^4 + 5x}{3x^3 - 4}$

- ☐ 0
- ☐ $-\frac{1}{3}$
- ☐ $-\infty$
- ☐ ∞
- ☐ $\frac{5}{3}$
- ☐ $-\frac{5}{4}$

2.4 Checking for understanding

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What is $\lim_{x \rightarrow -\infty} \frac{5x-3}{\sqrt{x^2+2}}$?

POSSIBLE POINTS: 1

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2.4 Checking for understanding

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POSSIBLE POINTS: 3

Find a vertical and horizontal asymptotes for $f(x) = \frac{x^2 - 2x}{25 - x^2}$

The horizontal asymptote is $y =$

The vertical asymptotes are $x =$ and $x =$

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Review

3.1 Checking for understanding

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Given function $y = x^2 + 3$.

a) The average rate of change of y with respect to x on the interval [2, 2.1] equals

b) The instantaneous rate of change of y with respect to x at the left endpoint of the interval equals

3.1 Checking for understanding

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A sandbag is dropped from a hot-air balloon that is hovering at a height of 600 feet above the ground.

If air resistance is disregarded, then the distance $s(t)$ from the ground to the sandbag after t second is given by: $-16t^2 + 600$. Find the velocity of the sandbag

a) at t sec ft/sec

b) at 2 sec ft/sec

3.1 Checking for understanding

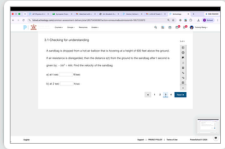
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Use limit definition of slope $m_a = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$ of the tangent line to the graph at the point with x -coordinate a to find the slope of the function $f(x) = \sqrt{x}$ at the point $P(9,3)$

- ☐ 1
- ☐ 1/6
- ☐ 9
- ☐ 1/9
- ☐ 6
- ☐ DNE

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Vertical toolbar with icons for grid, person, flag, pencil, calculator, ruler, protractor, and clipboard.



3.2 Checking for understanding

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If $y = -3x + 10$, then $\frac{d^3y}{dx^3} =$

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3.2 Checking for understanding

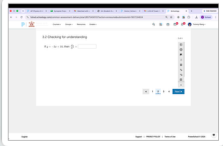
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$\lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h} = ?$

- ☐ x^4
- ☐ $(x + h)^4$
- ☐ $4x^3$
- ☐ $(4x^3)'$
- ☐ nonexistent

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3.2 Checking for understanding

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Given a function $\begin{cases} 3x, & \text{if } x \leq 0 \\ x^3, & \text{if } x > 0 \end{cases}$ Use the graph of f to find the domain of f'

- ☐ $(-\infty, +\infty)$
- ☐ $(-\infty, 3)$
- ☐ $(-\infty, 3) \cup (3, +\infty)$
- ☐ $(0, \infty)$
- ☐ $(-\infty, 0) \cup (0, +\infty)$

3.3 Checking for Understanding

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POSSIBLE POINTS: 1

Given $f(x) = \frac{3x-2}{5x+1}$, find $f'(2)$.

$f'(2) =$

Give exact answer as a fraction.

3.3 Checking for Understanding

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POSSIBLE POINTS: 1

Find the derivative: $h(x) = (x^2 - 4)(x^4 + 2)$

$h'(x) =$

- ☐ $(2x)(4x^3) + (4x^3)(2x)$
- ☐ $(x^2 - 4)(4x^3) - (x^4 + 2)(2x)$
- ☐ $(2x)(4x^3)$
- ☐ $(x^2 - 4)(4x^3) + (x^4 + 2)(2x)$

3.3 Checking for Understanding

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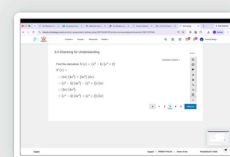
POSSIBLE POINTS: 2

If f and g are functions such that $f(3) = 2$, $f'(3) = 8$, $g(3) = -3$, and $g'(3) = 5$, then

a) $\left(\frac{1}{f}\right)'(3) = \boxed{}$

b) $(fg)'(3) =$

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3.3 Checking for Understanding

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POSSIBLE POINTS: 1

Find the point P on the graph of $y = x^2$ such that the tangent line at P has x-intercept 1. Give exact answer or round to three decimal places.

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3.4 Checking for Understanding

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POSSIBLE POINTS: 1.67

If $h(z) = \frac{1-\sin z}{1+\cos z}$, then $\frac{dh}{dz}$ for $z = 0$ equals

Give exact answer or rounded to three decimal places. If the the derivative is undefined, write DNE

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3.4 Checking for Understanding

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POSSIBLE POINTS: 1.67

$\lim_{x \rightarrow 0} \frac{\sin(2x)}{x} =$

Give exact answer or round to three decimal places.



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3.4 Checking for Understanding

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POSSIBLE POINTS: 1.67

If $r(\theta) = \cot \theta + \csc \theta$ then $r'(\frac{\pi}{6}) =$.

Give exact answer or round to three decimal places.

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3.6 Checking for Understanding

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If $m(t) = \tan^2(\sqrt{x^2 - 5})$, find $m'(3)$.

- ☐ $\tan(2) \sec^2(2)$
- ☐ $3 \tan(2) \sec^2(2)$
- ☐ $4 \tan(2) \sec(2)$
- ☐ $\frac{\sec^2(6)}{2}$
- ☐ $\frac{\tan(2) \sec^2(2)}{4}$

3.6 Checking for Understanding

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Use the equation of the tangent line to $f(x) = x^2(3 - x)$ at $x = 1$ to approximate $f(1.1)$.

$f(1.1) \approx$

Give exact answer or round to three decimal places.

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3.6 Checking for Understanding

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A particle moves along the curve $y = x^3 - 2$ such that $\frac{dx}{dt} = xy$. Find $\frac{dy}{dt}$ at the point $(x, y) = (2, 6)$.

$$\left[\frac{dy}{dt} \right]_{(2,6)} = \boxed{}$$

Give exact answer or round to three decimal places.

Review

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3.7 Checking for Understanding

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POSSIBLE POINTS: 1.67

Given the equation $x^2 - 2y^2 = 18$, find y' at the point $(6, 3)$.

$[y']_{(6,3)} =$

Give exact answer or round to three decimal places.

3.7 Checking for Understanding

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POSSIBLE POINTS: 1.67

Given the equation $xy = \sin y$, find y' at the point $(0, \pi)$.

$[y']_{(0,\pi)} =$

- ☐ 0
- ☐ 1
- ☐ -1
- ☐ $-\pi$
- ☐ π
- ☐ Does not exist

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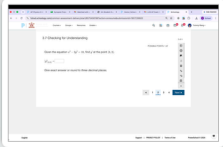
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3.7 Checking for Understanding

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POSSIBLE POINTS: 1.67

The equation $x^2 + y^2 + \sqrt{2}xy = 1$ has horizontal tangent lines at two points. Find the two points where the equation has horizontal tangent lines.

 and

Give exact answer or round to three decimal places.

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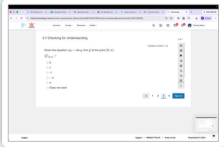
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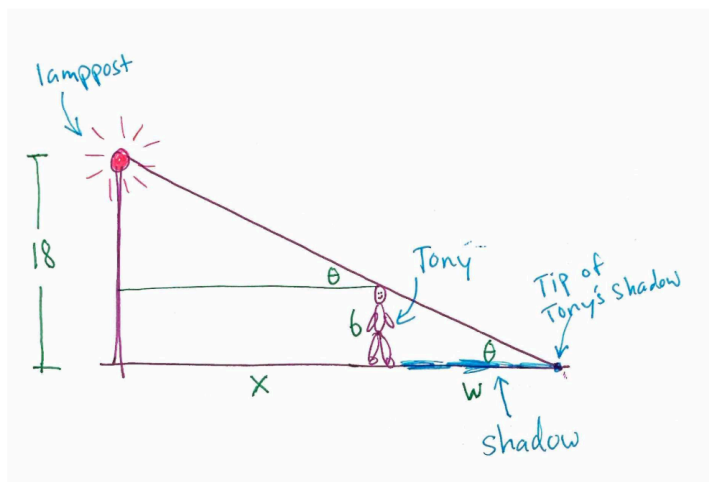
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3.8 Checking for Understanding

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Tony, a 6 ft tall person is walking away from an 18 ft tall lamppost at night at a rate of 3 ft/sec. The moment Tony's head is 24 ft from the top of the lamppost, find:

- The rate at which the length of Tony's shadow is increasing is ft/sec.
- The rate at which the tip of Tony's shadow is moving is ft/sec.
- The rate at which the acute angle formed by the ground and the line connecting the tip of the shadow and Tony's head is changing is radians/sec.

Give exact answers or round to three decimal places.

3.8 Checking for Understanding

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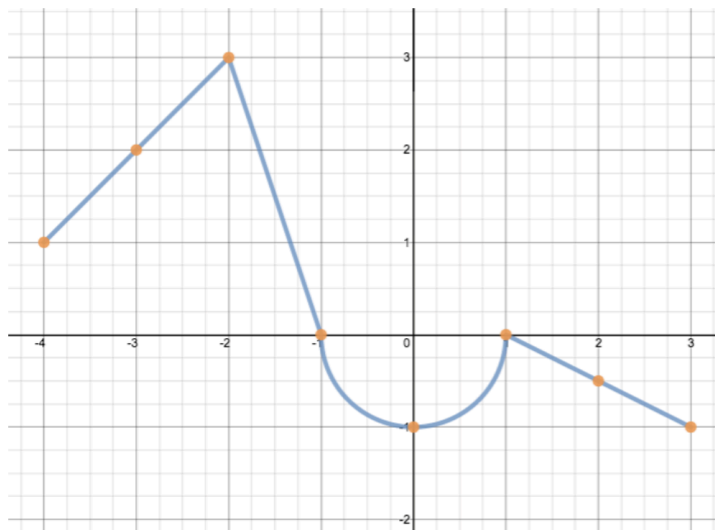
A hypothetical square has side lengths that are changing at a rate of 2 cm/sec. When the side lengths are 6 cm in length, find:

- a) the rate at which the diagonal is changing: cm/sec
- b) the rate at which the area of the square is changing: square cm / sec

Give exact answers or round to three decimal places.

POSSIBLE POINTS: 2

The graph of $y = f(x)$ is shown below.



- a) The maximum value of f is .
- b) How many critical numbers for f are there on the interval $(-4, 3)$? There are critical numbers for f on the interval $(-4, 3)$.
- c) f has a local maximum at the x-values and .

4.1 Checking for Understanding

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POSSIBLE POINTS: 1

The absolute maximum for $y = x^3 - 4x^2 - 5$ on the closed interval $[-2, 5]$ is

- ☐ 5
- ☐ 0
- ☐ -5
- ☐ 20
- ☐ 25

Navigation buttons: back, 1, 2, 3 (selected), 4, 5, Next

Vertical toolbar with icons for: grid, no calculator, flag, pencil, calculator, eraser, protractor, and clipboard.



4.1 Checking for Understanding

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POSSIBLE POINTS: 1

The function $y = x^2$ is increasing on the interval

The function $y = x^3$ is increasing on the interval

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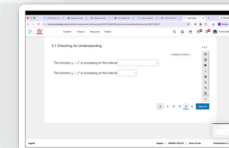
4.1 Checking for Understanding

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POSSIBLE POINTS: 1

Select all of the critical numbers for the function $y = \frac{x^2+2x-8}{x-4}$.

- ☐ 0
- ☐ 2
- ☐ 4
- ☐ 6
- ☐ 8
- ☐ 10



4.2 Checking for Understanding

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A function f has $f(2) = 5$ and $f(5) = 3$. Select all additional information needed to justify the claim that there exists a value c between 2 and 5 such that $f'(c) = -\frac{2}{3}$.

- ☐ f is continuous on $(2, 5)$
- ☐ f is continuous on $[2, 5]$
- ☐ f is differentiable on $(2, 5)$
- ☐ f is differentiable on $[2, 5]$

4.2 Checking for Understanding

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Match the below functions with the statement that best describes the function. While a given selection may describe multiple functions, there is only one way to match them all up that uses all of the possible answers.

$$y = |x - 1| \text{ for } -2 \leq x \leq 2$$

$$y = \begin{cases} x + 2 & -2 \leq x \leq 0 \\ -x + 2 & 0 < x \leq 2 \end{cases}$$

$$y = \begin{cases} x + 2 & -2 \leq x < 2 \\ 0 & x = 2 \end{cases}$$

$$y = x \text{ for } -2 \leq x \leq 2$$

⚡ This function does not satisfy the MVT because the function is not differentiable on $(-2, 2)$.

⚡ This function does not satisfy Rolle's Theorem because the function is not differentiable on $(-2, 2)$.

⚡ This function does not satisfy Rolle's Theorem because the function is discontinuous on $[-2, 2]$.

⚡ This function does not satisfy Rolle's Theorem because $f(-2) \neq f(2)$



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4.2 Checking for Understanding

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For the function $y = x^2$, the value(s) of x that satisfy the Mean Value Theorem on the interval $[1, 3]$ are

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4.3 Checking for Understanding

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POSSIBLE POINTS: 2.5

Find the x-values of any local extrema of f and the intervals on which f is increasing or decreasing.
Select all answers that apply.

$$f(x) = 4x^3 - 3x^4$$

- ☐ local max at $x=0$
- ☐ local min at $x=0$
- ☐ local max at $x=1$
- ☐ local min at $x=1$
- ☐ f is increasing on $(-\infty, 0) \cup (0, 1)$
- ☐ f is decreasing on $(1, \infty)$
- ☐ f is increasing on $[0, \frac{4}{3}]$
- ☐ f is decreasing on $(-\infty, 0] \cup [\frac{4}{3}, \infty)$
- ☐ local max at $4/3$

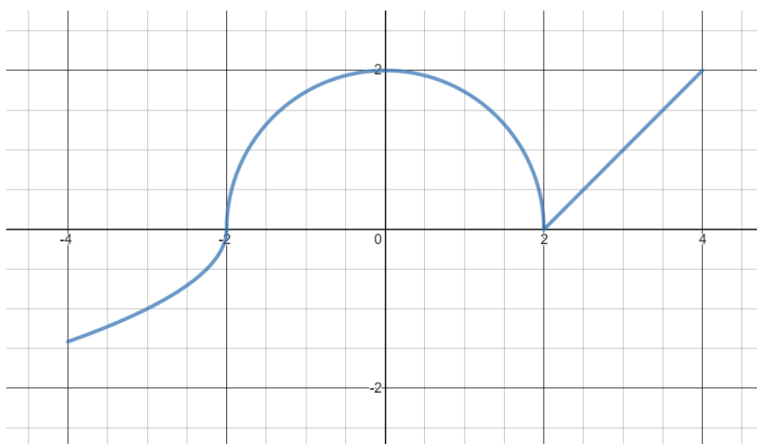


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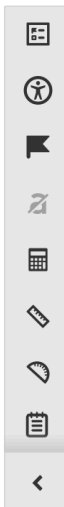
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POSSIBLE POINTS: 2.5

The graph of $y = f'(x)$ is shown below. This is the graph of the **derivative** of f . The function f is defined on the interval $(-4, 4)$.



The function f has a local at and no local .



4.4 Checking for Understanding

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POSSIBLE POINTS: 1.88

Use the table below to answer the following questions.

x	$f'(x)$	$f''(x)$
2	-2	0
3	0	0
4	0	3

a) T/F: Based on the table above, f must have a local maximum at $x = 4$. ☐

b) T/F: Based on the table above, f must have a point of inflection at $x = 2$. ☐

c) T/F: Based on the table above, f must have a point of inflection at $x = 3$. ☐

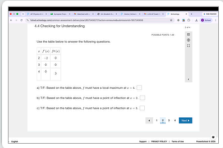
4.4 Checking for Understanding

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POSSIBLE POINTS: 1.88

The function $y = \frac{1}{20}x^5 - \frac{1}{3}x^4 + \frac{2}{3}x^3 + 70000$ has points of inflection at which values of x ? Select all that apply.

- ☐ -2
- ☐ -1
- ☐ 0
- ☐ 1
- ☐ 2

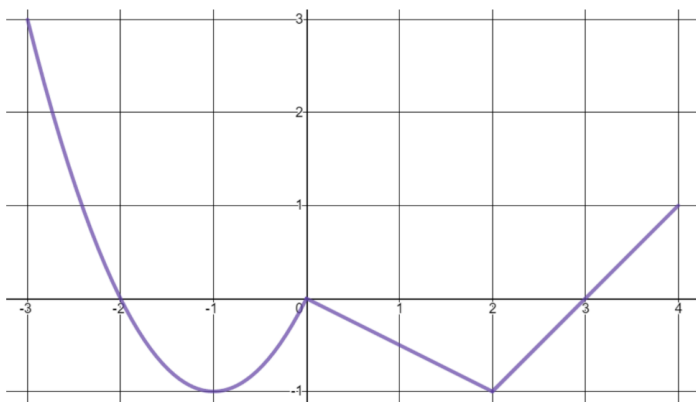


4.4 Checking for Understanding

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POSSIBLE POINTS: 1.25

The differentiable function $y = f(x)$ is defined on the interval $(-3, 4)$. The graph of $y = f'(x)$, the **derivative of f** is shown below.



a) $y = f(x)$ is concave up on the interval(s)

b) $y = f(x)$ has a point of inflection at the x-values



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